EE446-TinyML Lab 2

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Github: https://github.com/sdadhich04/EE-446-TInyML-

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

Raw audio in the time domain only represents amplitude variations over time, which does not explicitly capture the frequency characteristics of speech. However, spoken words are primarily distinguished by their frequency content and how those frequencies evolve over time. By converting the waveform into spectrogram or MFCC-style features, we obtain a structured time-frequency representation that highlights these patterns. This transformation makes it significantly easier for the model to learn discriminative features between keywords such as "yes" and "no," leading to better performance and more efficient training.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

The representative dataset is used during full integer quantization to calibrate the dynamic range of activations within the model. Since the model is being converted from floating-point to INT8, TensorFlow Lite needs to determine appropriate scaling factors and zero-points for each layer. The representative dataset provides sample inputs that reflect real-world data distributions, allowing the converter to accurately map floating-point values to integer ranges. Without proper calibration, quantization can introduce significant information loss, resulting in degraded model accuracy. Therefore, the representative dataset is essential for maintaining performance in the INT8 model.

3. What does audio_processor.get_data() do? Refer to input_data.py.

The audio_processor.get_data() function is responsible for retrieving batches of input data and corresponding labels from the dataset. It not only loads the audio samples but also applies the necessary preprocessing steps, such as feature extraction, to convert them into the format required by the model. In essence, it serves as the data pipeline that bridges raw dataset inputs and model-ready features during both training and evaluation.

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are g_model and g_model_len, and why do they need to be copied correctly into the Arduino source file?

In the final deployment step, the TensorFlow Lite model is converted into a C/C++ byte array. The variable g_model stores the serialized model as a byte array, while g_model_len specifies the size of this array in bytes. These variables are used by the TensorFlow Lite Micro interpreter on the Arduino to load and execute the model directly from memory. It is critical that both the byte array and its length are copied accurately into the Arduino source file, as any mismatch or corruption can prevent the model from loading correctly or lead to incorrect inference results.

5. Please include a screenshot showing the correct identification of at least one keyword, either “YES” or “NO”, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.

```
Heard unknown (206) @75776ms
Heard unknown (205) @78416ms
Heard unknown (202) @80224ms
Heard yes (219) @84752ms
Heard no (205) @86464ms
Heard yes (208) @89104ms
Heard no (204) @90624ms
Heard unknown (212) @91664ms
Heard unknown (204) @93168ms
Heard unknown (201) @94688ms
Heard unknown (202) @96288ms
Heard unknown (213) @100928ms
Heard unknown (222) @102448ms
Heard unknown (201) @104704ms
Heard unknown (202) @108112ms
Heard unknown (219) @109632ms
Heard unknown (230) @111136ms
Heard unknown (204) @113216ms
Heard unknown (202) @115776ms
Heard no (201) @117488ms
Heard unknown (207) @123344ms
Heard no (207) @124480ms
Heard no (202) @127216ms
Heard no (206) @129680ms
Heard unknown (203) @134304ms
Heard unknown (203) @138656ms
Heard unknown (218) @140176ms
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